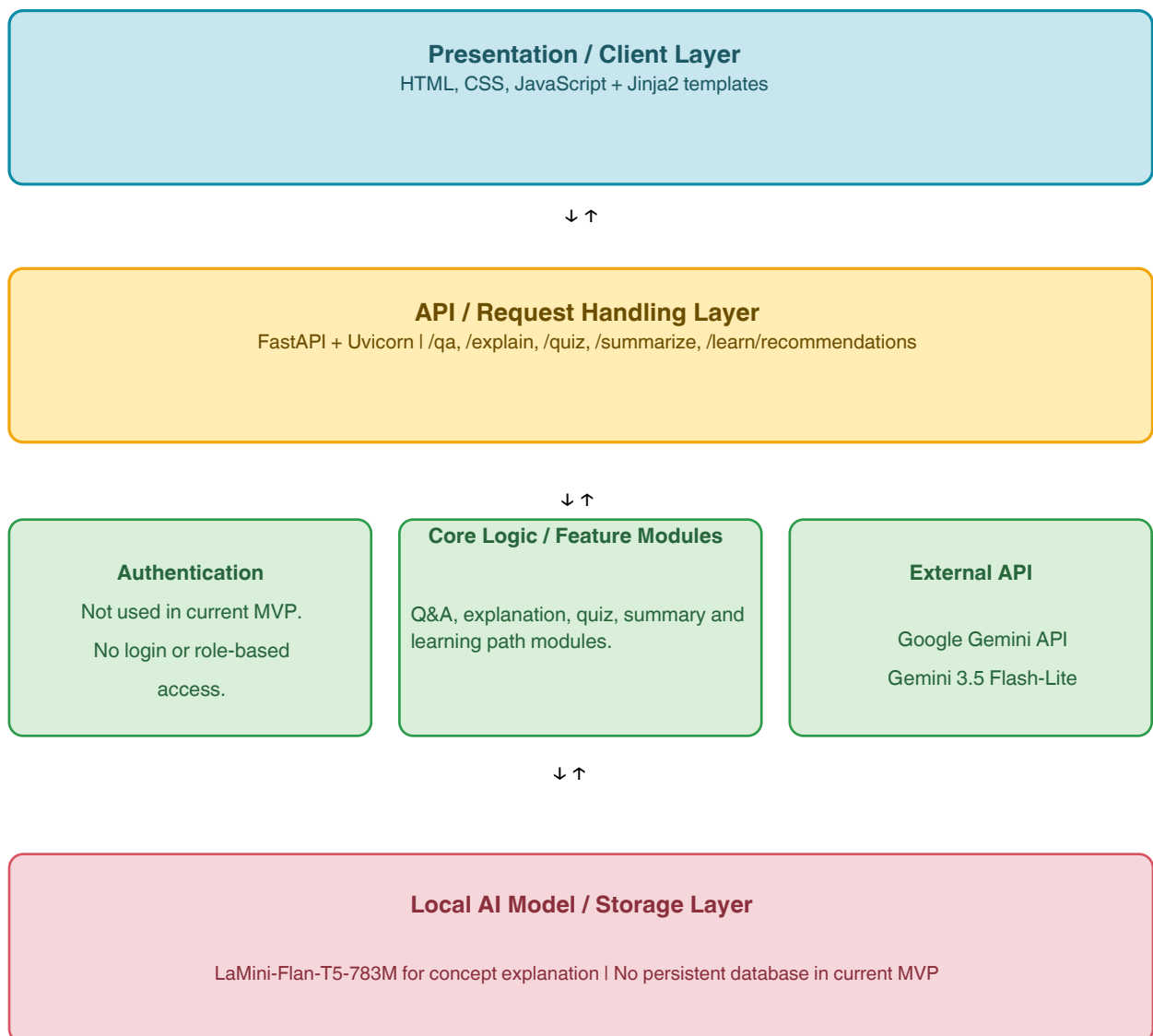


Solution Architecture

Date	30 September 2026
Team ID	6ab39579211fa4fd5d28901b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture of EduGenie showing the web interface, FastAPI request handling, feature modules, Gemini API integration, local concept explanation model, and the absence of a persistent database.



EduGenie uses Gemini for generative learning features and a local LaMini model for concept explanation.

	Component DescriptionTable	
API / Request Handling	Receives requests, validates inputs, route seachrequest to the appropriate feature module, and returns the generated result.	Python, FastAPI, Uvicorn, Pydantic
Feature Modules	Implements Q&A;, concept explanation, quiz generation, summarization, and personalized learning path generation.	Python, Gemini 3.5 Flash-Lite, LaMini-Flan-T5-783M
AI / External Integration	Uses Gemini for Q&A;, quizzes, summaries and learning paths. Concept explanation is handled locally by LaMini-Flan-T5-783M.	Google Gemini API, LaMini-Flan-T5-783M
Data / Storage	The current MVP does not use a persistent database. Learning requests and generated responses are processed during feature use.	No persistent database

Architecture Notes

- No separate authentication service is implemented in the current MVP.
- No persistent database is used.
- Gemini API access uses the API key from environment configuration; the key is not stored in source code.
- The application runs locally through Uvicorn for development, testing and demonstration.